VANISHING OF HIGHER CECH COHOMOLOGY FOR ALL OPEN COVERINGS IMPLIES VANISHING OF COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: it had been excluded only because the chapter's retained preamble lacks the slogan environment. The author's slogan and the Xy-pic diagram of the proof are both part of the counted spans; the card carries a per-card preamble copy formed from the chapter retained preamble plus the project's own three display environments copied verbatim from sources/stacks-chow/preamble.tex lines 7-12.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The label lemma-cech-vanish is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card. The variant of this lemma stated on a basis of opens is a different statement counted by its own card.

Allowed standard prerequisites (editorial).

- Injective modules have trivial higher Cech cohomology. Same chapter, Lemma lemma-injective-trivial-cech, invoked at proof lines 1652-1653. Supplies the vanishing for the injective used in the induction.
- The hypothesis upgrades a short exact sequence of sheaves to a short exact sequence of presheaves. Same chapter, Lemma lemma-ses-cech-h1, invoked at proof lines 1659-1660. Supplies the exactness of the sequence of presheaves and hence the Cech vanishing of Q.
- Cech cohomology is a cohomological functor on presheaves of modules. Same chapter, Lemma lemma-cech-cohomology-delta-functor-presheaves, invoked at proof lines 1661-1663. Supplies the long exact sequence used in the induction.
- Higher cohomology of an injective module vanishes. Derived Categories, Lemma derived-lemma-higher-derived-functors, invoked at proof lines 1683-1685. Supplies H^n(U, I) = 0 for n > 0.

Source reference. Original lemma 015B: sources/stacks-chow/tags.txt, source lines 1502–1502. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EK: sources/stacks-cohomology/pinned-original-source.tex, source lines 1066–1066. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EP: sources/stacks-cohomology/pinned-original-source.tex, source lines 1408–1408. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EU: sources/stacks-cohomology/pinned-original-source.tex, source lines 1593–1593. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module such that*

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$$

for all $p > 0$ and any open covering $\mathcal{U} : U = \bigcup_{i \in I} U_i$ of an open of X. Then $H^p(U, \mathcal{F}) = 0$ for all $p > 0$ and any open $U \subset X$.

Proof. Let $\mathcal{F}$ be a sheaf satisfying the assumption of the lemma. We will indicate this by saying "$\mathcal{F}$ has vanishing higher Čech cohomology for any open covering". Choose an embedding $\mathcal{F} \to \mathcal{I}$ into an injective $\mathcal{O}_X$-module. By Lemma 01EP $\mathcal{I}$ has vanishing higher Čech cohomology for any open covering. Let $\mathcal{Q} = \mathcal{I}/\mathcal{F}$ so that we have a short exact sequence

$$0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0.$$

By Lemma 01EU and our assumptions this sequence is actually exact as a sequence of presheaves! In particular we have a long exact sequence of Čech cohomology groups for any open covering $\mathcal{U}$, see Lemma 01EK for example. This implies that $\mathcal{Q}$ is also an $\mathcal{O}_X$-module with vanishing higher Čech cohomology for all open coverings.

Next, we look at the long exact cohomology sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(U, \mathcal{F}) & \longrightarrow & H^0(U, \mathcal{I}) & \longrightarrow & H^0(U, \mathcal{Q}) \\ & & & \searrow & & & \\ & & H^1(U, \mathcal{F}) & \longrightarrow & H^1(U, \mathcal{I}) & \longrightarrow & H^1(U, \mathcal{Q}) \\ & & & \searrow & & & \\ & & \ldots & & \ldots & & \ldots \end{array}$$

for any open $U \subset X$. Since $\mathcal{I}$ is injective we have $H^n(U, \mathcal{I}) = 0$ for $n > 0$ (see Derived Categories, Lemma 015B). By the above we see that $H^0(U, \mathcal{I}) \to H^0(U, \mathcal{Q})$ is surjective and hence $H^1(U, \mathcal{F}) = 0$. Since $\mathcal{F}$ was an arbitrary $\mathcal{O}_X$-module with vanishing higher Čech cohomology we conclude that also $H^1(U, \mathcal{Q}) = 0$ since $\mathcal{Q}$ is another of these sheaves (see above). By the long exact sequence this in turn implies that $H^2(U, \mathcal{F}) = 0$. And so on and so forth. □